

In this paper, without the explain,  $\|\cdot\|$  means  $\|\cdot\|_2$ , the 2-norm.

$$A = \begin{pmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & & & \\ \vdots & & & \\ A_{n1} & \cdots & \cdots & A_{nn} \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} B_{11} & B_{12} & \cdots & B_{1n} \\ B_{21} & & & \\ \vdots & & & \\ B_{n1} & \cdots & \cdots & B_{nn} \end{pmatrix}$$

Denote  $e_i \in F^{n \times 1}$ , then

$$Ae_i = \begin{pmatrix} A_{1i} \\ A_{2i} \\ \vdots \\ A_{ni} \end{pmatrix} \quad \text{and} \quad e_i^T A = (A_{i1} \ A_{i2} \ \cdots \ A_{in})$$

$$AB = \left( \sum_{k=1}^n A_{ik} \cdot B_{kj} \right)_{i,j} = \left( f(A^T e_i, A e_j) \right)_{i,j},$$

where  $f: F^n \times F^n \rightarrow F: \left( \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \right) \mapsto \sum_{k=1}^n x_k y_k$ , which is the bilinear form on  $F^n$ .

$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}_{n \times 1} \cdot \begin{bmatrix} b_1 & b_2 & \cdots & b_n \end{bmatrix}_{1 \times n} = \begin{bmatrix} a_1 b_1 & a_1 b_2 & \cdots & a_1 b_n \\ a_2 b_1 & & & \\ \vdots & & a_{ij} b_j & \\ \vdots & & & \\ a_n b_1 & - & - & a_n b_n \end{bmatrix}$$

# Basic Matrix

Def. Let  $A \in M_{m \times n}(F)$  be given where  $F = \mathbb{R}$  or  $\mathbb{C}$ . Define the supremum norm of  $A$

$$\|A\|_p \stackrel{\text{def}}{=} \sup_{x \neq 0} \frac{\|Ax\|_p}{\|x\|_p} \stackrel{(*)}{=} \sup_{\|x\|_p=1} \|Ax\|_p$$

where  $\|\cdot\|_p$  is the  $p$ -norm, that is,

$$\|\cdot\|_p: F^n \rightarrow \mathbb{R}; (a_i)_{i=1}^n \mapsto \left( \sum_{i=1}^n |a_i|^p \right)^{\frac{1}{p}}.$$

The equality  $(*)$  is given by the fact that for any non-zero vector  $x$ , then

$$\frac{\|Ax\|_p}{\|x\|_p} = \left\| A \cdot \frac{x}{\|x\|_p} \right\|_p = \|Ax'\|_p, \text{ where } \|x'\|_p = 1.$$

Prop. Given  $A \in M_{m \times n}(F)$ , the following conditions are satisfied:

$$1) \|A\|_1 = \max_{1 \leq j \leq n} \|Ae_j\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^m |A_{ij}|$$

$$2) \|A\|_\infty = \max_{1 \leq i \leq m} \|e_i^T A\|_1 = \max_{1 \leq i \leq m} \sum_{j=1}^n |A_{ij}|$$

$$3) \|A\|_2 = \sigma_{\max}(A), \text{ the largest singular value for } A.$$

Proof. By definition,  $\|A\|_p = \sup_{\|x\|_p=1} \|Ax\|_p$  and given  $x \in F^n$  s.t.  $\|x\|_p = 1$ , ( $x = (x_1, \dots, x_n)$ )

$$\|Ax\|_1 = \left\| \sum_{k=1}^n x_k Ae_k \right\|_1 \leq \sum_{k=1}^n |x_k| \cdot \|Ae_k\|_1 \leq \sum_{k=1}^n |x_k| \cdot \max_{1 \leq j \leq n} \|Ae_j\|_1 = \max_{1 \leq j \leq n} \|Ae_j\|_1$$

And, take  $x = e_j$  where the index  $1 \leq j \leq n$  satisfying  $\|Ae_j\|_1$  is the maximal value, we obtain the equality. 1) And, to show the second assertion, observe that

$$\|Ax\|_\infty = \left\| \sum_{k=1}^n x_k Ae_k \right\|_\infty \leq \sum_{k=1}^n |x_k| \cdot \|Ae_k\|_\infty$$

## Differentiation.

Def. Let  $A \subseteq \mathbb{R}^n$ ; let  $f: A \rightarrow \mathbb{R}^m$ . For  $a \in \text{int } A$  and  $u \in \mathbb{R}^n$  with  $u \neq 0$ , the limit

$$f'(a; u) \stackrel{\text{def}}{=} \lim_{t \rightarrow 0} \frac{f(a+tu) - f(a)}{t} \quad (\text{if the limit exists})$$

is called the directional derivative of  $f$  at  $a$  with respect to  $u$ .

And,  $f$  is said to be differentiable at  $a$  if there is a linear map  $T \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$  s.t.

$$\frac{f(a+h) - f(a) - T(h)}{\|h\|} \rightarrow 0 \quad \text{as } h \rightarrow 0. \quad (h \in \mathbb{R}^n)$$

Such a linear map  $T$  is unique, and it is denoted  $Df_a = T$ , called derivative.

Thm. Let  $A \subseteq \mathbb{R}^n$ ; let  $f: A \rightarrow \mathbb{R}^m$ .

If  $f$  is differentiable at  $a$ , then all the directional derivatives of  $f$  at  $a$  exist satisfying  $f'(a; u) = Df(a) \cdot u$  for all  $u \in \mathbb{R}^n$ .

Proof By definition, given  $u \in \mathbb{R}^n$  with  $u \neq 0$ ,

$$\lim_{t \rightarrow 0} \frac{f(a+tu) - f(a) - Df(a) \cdot (tu)}{\|t \cdot u\|} = 0.$$

Meanwhile, since  $Df(a)$  is linear,  $Df(a) \cdot (tu) = t \cdot Df(a) \cdot u$ .

$$\begin{aligned} \lim_{t \rightarrow 0} \left\| \frac{f(a+tu) - f(a)}{\|t\| \cdot \|u\|} - \frac{t \cdot Df(a) \cdot u}{\|t\| \cdot \|u\|} \right\| \\ = \lim_{t \rightarrow 0} \left\| \frac{f(a+tu) - f(a)}{t} - Df(a) \cdot u \right\| \cdot \frac{|t| \cdot \|u\|}{\|t\| \cdot \|u\|} = 0 \end{aligned}$$

So proved.  $\square$

But, the converse is not true: observe

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}; (x, y) \mapsto \frac{x^2 y}{x^2 + y^2} \quad \text{if } (x, y) \neq (0, 0), \text{ but } (0, 0).$$

Def. Given  $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ , define the  $j$ th partial derivative at  $a$  as

$$D_j f(a) \stackrel{\text{def}}{=} f'(a, e_j).$$

Thm. Given  $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ , denote  $f_i = \pi \circ f: \mathbb{R}^n \rightarrow \mathbb{R}$ . ( $1 \leq i \leq m$ )

If  $f_i$  is differentiable at  $a$ , then

$$Df_i(a) = [D_1 f_i(a) \quad D_2 f_i(a) \quad \dots \quad D_n f_i(a)]$$

Moreover,

$f$  is differentiable at  $a$  if and only if for each  $f_i$  is differentiable at  $a$ .

Proof. The first assertion is directly, by

$$Df_i(a) \cdot e_j \stackrel{\text{Lemma}}{\equiv} f_i(a; e_j) \stackrel{\text{Def.}}{\equiv} D_j f_i(a)$$

And, observe that: let  $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$  be any linear map. Consider a map

$$F(h) = \frac{f(a+h) - f(a) - T \cdot h}{\|h\|} \text{ for small enough } h \in \mathbb{R}^n.$$

Since the projection is linear,

$$\pi_i \circ F(h) = \frac{f_i(a+h) - f_i(a) - \pi_i \circ T \cdot h}{\|h\|}$$

By the topological property, the limit  $\lim_{h \rightarrow 0} F(h)$  exists iff  $\pi_i \circ F(h) \rightarrow 0$  for each  $i$ .

Thus the first assertion gives that the result. Visually,

$$Df(a) = (D_j f_i(a))_{i,j} = \begin{bmatrix} D_1 f_1(a) & D_2 f_1(a) & \dots & D_n f_1(a) \\ D_1 f_2(a) & & & \vdots \\ \vdots & & \ddots & \vdots \\ D_1 f_m(a) & \dots & \dots & D_n f_m(a) \end{bmatrix}$$

$$\left( = \frac{\partial f_i}{\partial x_j}(a) \right)$$

Thm. Let  $A$  be an open subset in  $\mathbb{R}^n$ , and let  $f: A(\subseteq \mathbb{R}^n) \rightarrow \mathbb{R}^3$ .  
 Suppose that for all  $1 \leq i \leq m$ ,  $1 \leq j \leq n$ , the partial derivatives  $D_j f_i$  exists and continuous.  
 Then,  $f$  is differentiable. (Such a function is called of class  $C^1$ ).

Proof. Since  $f$  is differentiable iff for each  $i$ ,  $f_i$  is differentiable,  
 so given  $1 \leq i \leq m$ , showing  $f_i$  is differentiable under the condition is enough.  
 Let  $a \in A$  be given. Given  $h = (h_1, \dots, h_n) \in \mathbb{R}^n$ , Consider

$$p_0 = a$$

$$p_1 = a + h_1 \cdot e_1$$

$$p_2 = a + h_1 \cdot e_1 + h_2 \cdot e_2$$

$$\vdots$$

$$p_n = a + h_1 e_1 + \dots + h_n e_n = a + h.$$

Then, by the telescoping,

$$f_i(a+h) - f_i(a) = \sum_{j=1}^n [f_i(p_j) - f_i(p_{j-1})]$$

And, for fixed  $1 \leq j \leq n$ ,

$$f_i(p_j) - f_i(p_{j-1}) = f_i(p_{j-1} + h_j e_j) - f_i(p_{j-1}) = D_j f_i(p_{j-1} + g_j e_j) \cdot h_j$$

where for some  $g_j$ , between 0 and  $h_j$ , by the Mean-Value Theorem.

(If  $h_j = 0$ , still the equality holds.). So,

$$f_i(a+h) - f_i(a) = \sum_{j=1}^n D_j f_i(p_{j-1} + g_j e_j) \cdot h_j.$$

And, since for  $B = [D_1 f_i(a) \quad \dots \quad D_n f_i(a)]$ ,

$$\frac{f_i(a+h) - f_i(a) - Bh}{\|h\|} = \frac{\sum_{j=1}^n [D_j f_i(p_{j-1} + g_j e_j) - D_j f_i(a)] \cdot h_j}{\|h\|} \rightarrow 0 \text{ as } h \rightarrow 0.$$

[Chain Rule.

[Thm. Let  $A \subseteq \mathbb{R}^n$  and  $B \subseteq \mathbb{R}^m$  be open and  $f: A \rightarrow \mathbb{R}^m$  and  $g: B \rightarrow \mathbb{R}^p$  with  $f[A] \subseteq B$ . If  $f$  is differentiable at  $a \in A$  and  $g$  is differentiable at  $f(a) \in B$ , then the composite  $g \circ f$  is differentiable at  $a$  with

$$D(g \circ f)(a) = Dg(b) \cdot Df(a).$$

Prat. Since  $B$  is open, choose  $\epsilon > 0$  such that  $B_\epsilon(f(a)) \subseteq B$ .

And, since  $f$  is continuous at  $a$ , there is  $\delta > 0$  such that

$$f[B_\delta(a)] \subseteq B_\epsilon(f(a))$$

In other words, given small enough  $\epsilon > 0$ , there is  $\delta > 0$  such that

$$\|x - a\| < \delta \Rightarrow \|f(x) - f(a)\| < \epsilon.$$

So that, the composition  $g \circ f$  is well-defined on  $B_\delta(a) \subseteq A$ .

Let  $\Delta(h) = f(a+h) - f(a)$  for  $\|h\| < \delta$ . And, define

$$F(h) = \frac{\Delta(h) - Df(a) \cdot h}{\|h\|} \quad \text{for } 0 < \|h\| < \delta.$$

Since  $f$  is differentiable at  $a$ , by definition,  $F(h) \rightarrow 0$  as  $h \rightarrow 0$ . So, since observing that

$$\|\Delta(h)\| = \|Df(a) \cdot h + \|h\| \cdot F(h)\| \leq \|Df(a)\| \cdot \|h\| + \|F(h)\| \cdot \|h\| = [\|Df(a)\| + \|F(h)\|] \cdot \|h\|.$$

Since  $F(h)$  converges, the quotient  $\|\Delta(h)\|/\|h\|$  is bounded in  $0 < \|h\| < \delta$ .

Now, define  $G$  as  $G(0) = 0$  and

$$G(k) = \frac{g(f(a) + k) - g(f(a)) - Dg(b) \cdot k}{\|k\|} \quad \text{for } 0 < \|k\| < \epsilon.$$

So that for all  $\|k\| < \epsilon$ ,

$$g(f(a) + k) - g(f(a)) = Dg(b) \cdot k + \|k\| \cdot G(k).$$

Consequently, substituting  $k = \Delta(h)$  and by the fact that  $f(a+h) = f(a) + \Delta(h)$ ,

$$\frac{1}{\|h\|} \cdot [g(f(a+h)) - g(f(a)) - Dg(b) \cdot Df(a) \cdot h] = \frac{1}{\|h\|} \cdot [Dg(b) \cdot h$$